27

Application of Process Integration Methodologies in the Brewing Industry

FRANÇOIS MARECHAL, ANURAG KUMAR SACHAN and LEANDRO SALGUEIRO, Ecole Polytechnique Fédérale de Lausanne (EPFL), Switzerland

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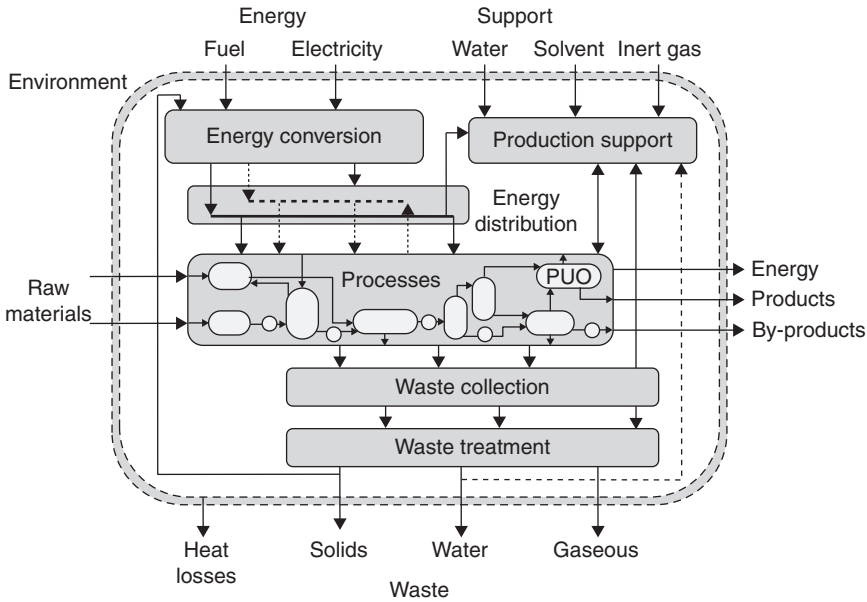
Abstract: This chapter presents the application of Process Integration techniques to increasing the energy efficiency of a typical brewery process. The example has been chosen to illustrate the application of Process Integration methodology and also to demonstrate the steps towards the implementation of Process Integration analysis. The chapter starts with a discussion on the definition of the process requirements, and of the system boundaries of the study. The Heat Cascade is then used to calculate the maximum heat recovery in the process. The integration of the refrigeration cycles and of the energy conversion technologies will then be studied to finally analyse the possible integration of heat pumping technologies as well as the integration of waste conversion units. It has been discussed how the targeting energy consumptions of the process can then be reached in a real industrial process implementation. The data used for the study are based on a flowsheeting model, and can therefore be considered as generic.

Key words: Process Integration, energy efficiency, Pinch Analysis, brewery process, cogeneration, heat pump, thermo-economic optimisation.

27.1 Introduction

Any industrial process is a system (Fig. 27.1) in which raw materials are converted into products and by-products. In most cases, the conversion is realised in a set of process unit operations or processing steps that require a production support such as water or solvent. The driving force of the conversion step is energy, which enters the system as energy resource flows being converted into a useful energy form such as heat, cold or work, in order to realise the required processing steps. As the conversion of the raw materials, as well as of the energy resources, is not perfect, part of the mass and the energy entering the system leaves the process as waste in the form of solid, liquid or gaseous, or as waste heat, which

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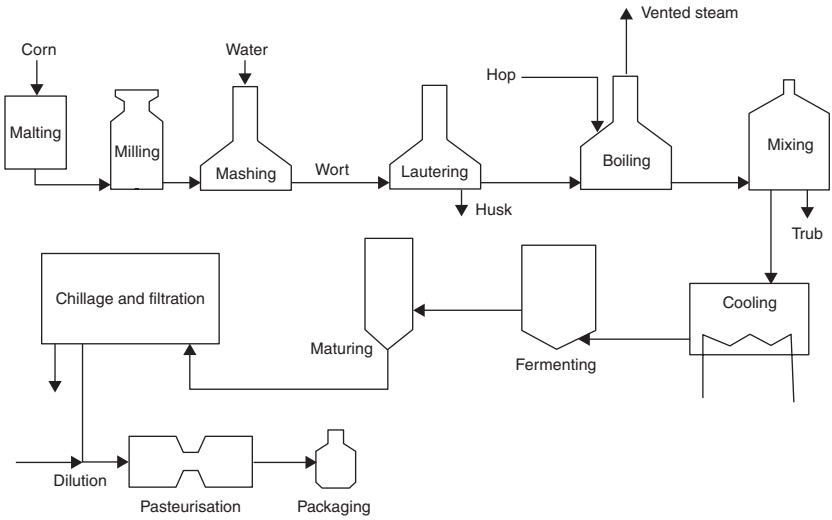


27.1 The industrial production system.

has to be treated and conditioned before leaving the system and released into the environment. When considering Process Integration, the goal is to optimise the overall efficiency of the process, thereby maximising the horizontal flows while minimising the vertical ones. Considering the system boundaries, Process Integration will address not only the processing steps, maximising the process throughput, but also the support streams management, maximising the recycling and the reuse of materials as well as the waste management and treatment and the energy conversion. This results sometimes in the addition of new products in the form of by-products or energy services.

The process used in this chapter is a beer-production process. The block flow diagram describing the major processing steps is sketched on Fig. 27.2. The beer production follows a recipe that converts a mixture of corn, water and hops (raw materials) into bottled beer (product) by means of a set of cooking, filtering and fermentation steps (processing steps). The process uses water as the production support, both for the process itself and for the equipment and conditioning washing. The conversion process requires energy in the form of heat for the cooking, and cold during the fermentation and conditioning. Waste resulting from the process is more or less humid, leaving the system as solid or liquid. In addition, the evaporation of a part of the water during the process also produces vapour, which is typically vented from the roof.

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27.2 The brewery process block flow diagram.

In the brewery, beer is produced from a mixture of water and malted corn. By volume, water is the main raw material. The different operations are now described in more detail, in order to characterise the use of energy in the beer processing.

Malting: In this process, the barley grains are prepared for brewing by releasing the starches in the barley. In a three-step approach, the grain is added to a vat with water and allowed to soak for approximately 40 h during steeping. Then, in germination, the grain is spread out on the floor for around five days and, finally, the green malt goes through very high temperature drying in the kilning step.

Milling: In the process of milling, the grains are cracked so that they can absorb the water. The water, as a result of this mixing, extracts sugar from the malt.

Mashing: The milled barley grains and supplementary grains (corn, sorghum, rye and wheat) are mixed with water and then heated to 65°C in a vessel for approximately 2 h in the process of mashing. During the mashing, the process of saccharification takes place with the mash, resulting into a sugar rich liquid, 'Wort'.

Lautering: This wort is filtered from the mash, in the 'lautering' process.

The remaining solid particles of the malt (brewing dregs) may be used as animal feedstock or biomethanised.

Boiling: The wort is then boiled at 105°C (under atmospheric pressure) for 1 h in a tank. During boiling, hops are added to control the bitterness, flavour and aroma; finally the mixture is sterilised. Approximately 10% by

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mass of the wort evaporates during the boiling, representing most of the total mass of wort that is evaporated in the brewing process.

Mixing–Whirlpool: The wort is then stirred and settles in a large tank, before being purified of all solid protein particles (trub). Approximately 0.2% by mass of the wort is settled and removed in the form of trub.

Cooling: This is the last step of the hot part of the brewing process. The purified wort is cooled to 10°C to prepare it for the process of fermentation.

Fermentation: Fermentation is the conversion of sugars to alcohols and carbon dioxide or organic acids, using yeasts or bacteria under anaerobic conditions. For this purpose the wort is stored in fermentation tanks, and maintained at a constant temperature of 10°C for 14 days. These chemical reactions are exothermic and, thus, cooling is necessary during fermentation.

Maturing: The wort is then maintained at 6°C for a period of 10 days for maturing. In this period, the CO_2 concentration is controlled and kept constant.

Chillage and Filtration: The wort is finally rapidly chilled to 1°C, and the yeast is eliminated by filtration. At this point a concentrated beer is obtained.

Dilution: The degasified cold water is mixed with the concentrated beer to achieve the desired alcohol content in the beer. In our case, it is diluted by adding 25% water by mass to the concentrated beer.

Pasteurisation: The beer is pasteurised by heating up to 70°C and, after a stay, cooling the beer to 5°C; it is then stored in the insulated tanks to be decanted into bottles or kegs.

Bottling: After pasteurisation, the beer is bottled. This step requires the bottle or kegs to be washed, and the packaging material to be prepared.

27.2 Process Flowsheet Analysis

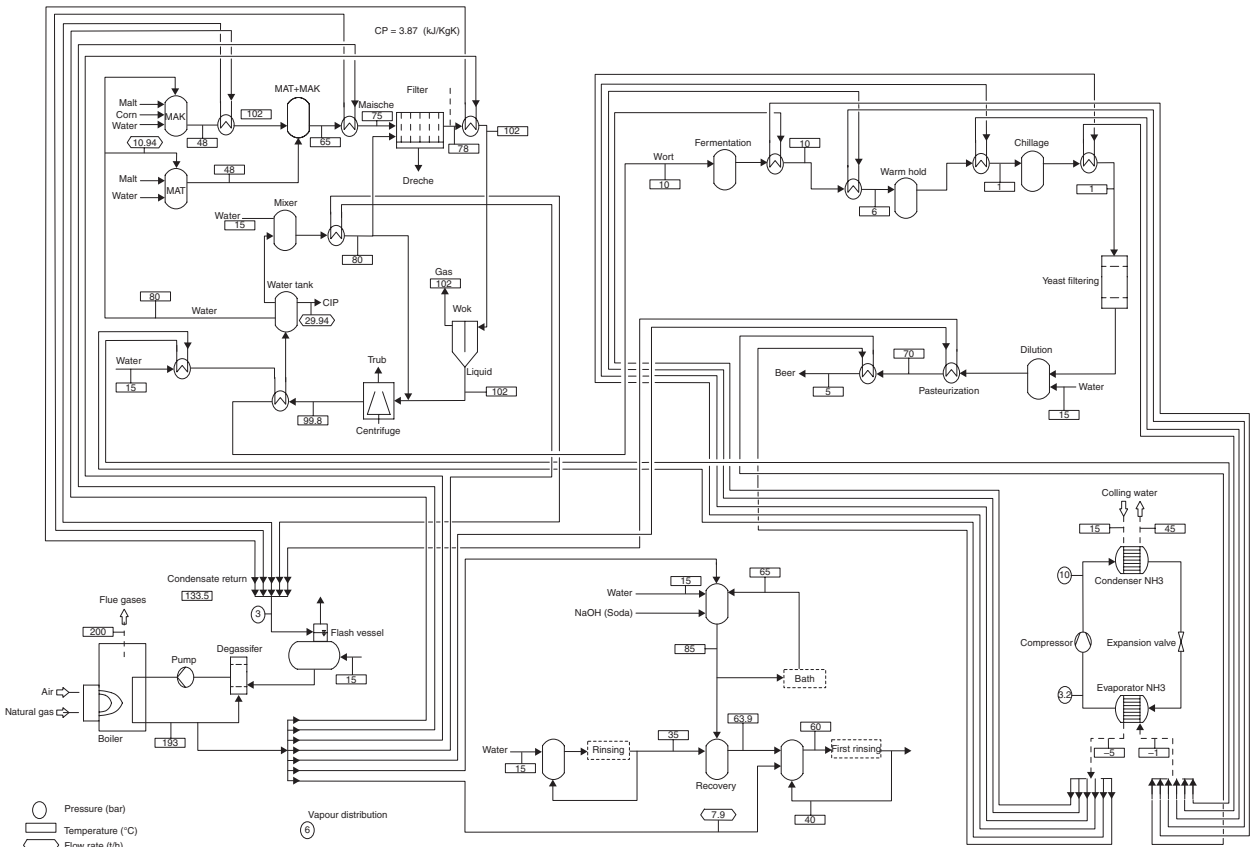
For defining the global system requirements, it is important to develop a systematic methodology that guarantees that all the needs and the boundaries of the system are well characterised. Using the brewery case as reference, in the next sections this methodology is described in detail.

27.2.1 Block Flow Process Analysis

The Flowsheet of Fig. 27.3 shows the flow of various streams within the brewing plant with constraints of temperature and pressure as per the recipe of the beer. The conventional plants do not have direct process/process heat recovery and thus heat exchangers exist only between process streams and the utility streams, including the hot water system.

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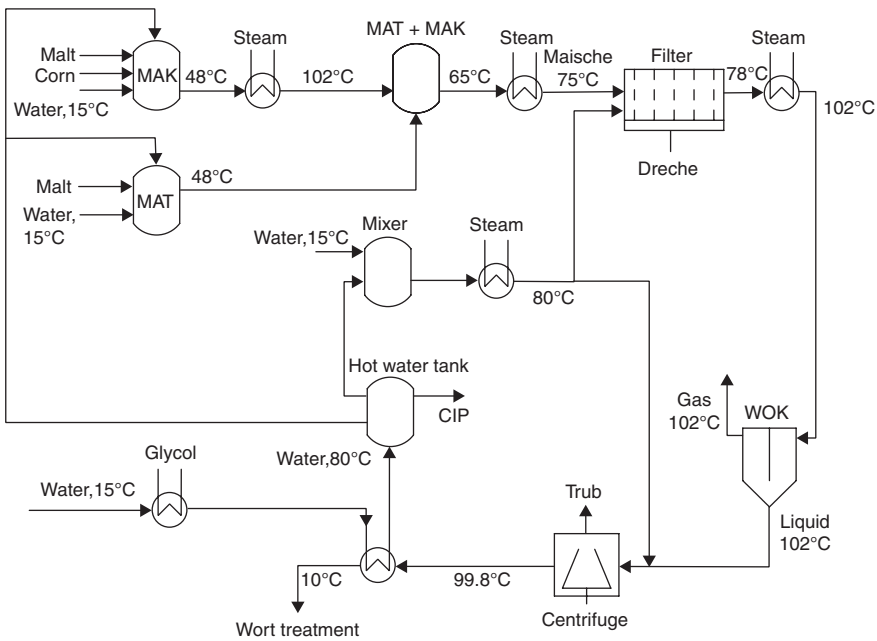


27.3 The Process Flowsheet.

The Process Flowsheet is divided into different sub-systems that are then allocated to process and utility sub-systems. The process subsystem refers to process units that are strictly related to process unit operations. The units in this system define therefore the requirements of the process. The utility system includes the units that are related to the supply of the energy (heating and cooling) services needed by the process. The utility subsystem includes the flows that we would like to optimise in order to minimise the energy bill.

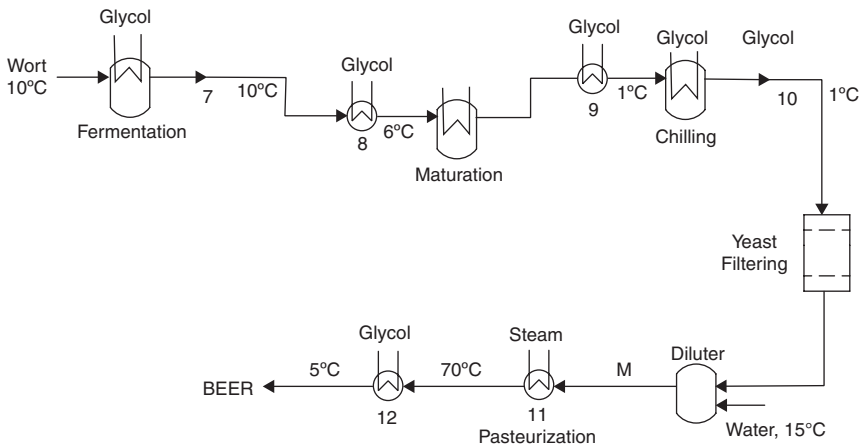
The process subsystem is defined into three major sub-systems:

Hot section: The hot part (Fig. 27.4) corresponds to the cooking part of the brewing process. The malt is first mixed with water in a large tank (MAT) and heated to a temperature of 48°C with the addition of hot water. The malt is especially made of sprouted barley. When malt is heated, the enzymes decompose the starch into sugar. Later, this sugar will be transformed into alcohol during fermentation. Another tank (MAK) filled with malt, corn and water is also preheated to 48°C; the resulting flow is then heated to a temperature of 102°C, before being mixed with the MAT tank content and reheated to 75°C. At this point, the mixture is called mash. This mash passes through a pressing filter, which requires an additional amount



27.4 Brewing process: Hot section.

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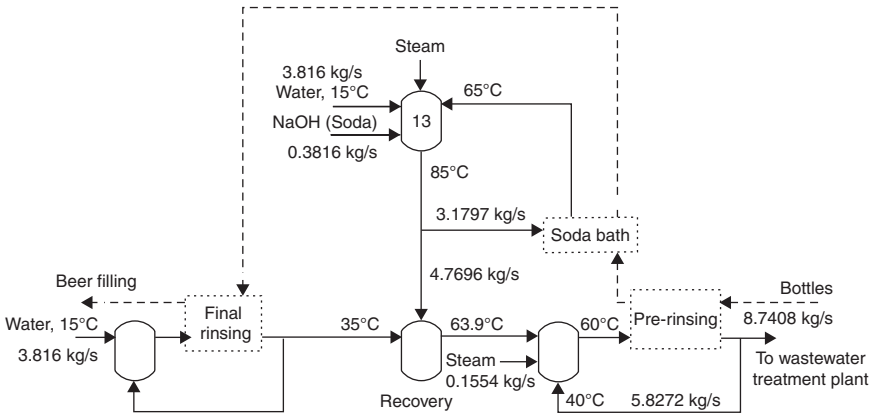


27.5 Brewing process: Cold section.

of hot water at 80°C. The wastage is essentially compressed malt that can be considered as dry and called dreche. After being filtered, the liquid is mixed with hops and heated in the WOK to a temperature of 102°C for approximately one hour. It is assumed that during this time about 10% of the liquid mass flow is evaporated. While the vapour is vented, the outgoing liquid (wort) is mixed with an additional amount of hot water before being filtered by a centrifugal device to separate the 0.2% of cooked proteins (trub). The wort is then cooled down to 10°C in a heat exchanger to reach the cold section of the process. The heat recovered is sent to the hot water tank to feed the hot water needs of the next operation.

Cold section: The cold section (Fig. 27.5) concerns mainly the fermentation and the subsequent beer-conditioning operations. The wort is first fermented at a temperature of 10°C. The fermentation reaction is exothermic; the observed heat of fermentation is 37.14 kJ/L_{wort}, and it includes heat gains from the ambience. Glycol–water is circulated in the fermentation tanks to maintain the temperature at 10°C for a fermentation period of 14 days. The wort is then cooled and maintained to a temperature of 6°C, to eliminate diacetyl residues that give the beer an unwanted taste of butter. This maturation period is usually 10 days, depending on the alcohol content required in the beer. The specific cooling requirement of the maturation is 29.55 kJ/L_{wort}. After the maturation process, the fresh beer is cooled down to 1°C, in order to precipitate the yeast that is filtered in the Chillage step. The specific heat for this step is 29.70 kJ/L_{beer}. The CO_2 concentration maintained during the maturation period is between 2.2 and 2.4 kg_{CO2} /hL_{wort}. At the end of the process, the alcohol concentration is controlled by diluting the beer mixture with water at 15°C. The beer is finally pasteurised, by heating up the mixture to 70°C and cooling it again to 5°C rapidly. This beer is now ready for packaging.

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27.6 Brewing process: Cleaning in place section.

Washing: The washing system (Fig. 27.6) mainly aims at preparing the packaging materials (mainly bottles and kegs) before filling them with beer. This process is considered as a continuous operation. The bottles are successively pre-rinsed, then washed with soda and rinsed in successive baths again, before being filled with beer. The temperature of the baths is maintained by steam injection. The final rinsing is done with clear water at 15°C. Water of the final bath rinsing is recuperated and sent to the pre-rinsing bath, recovering heat from the bottles that are cooled down in the last bath.

It is assumed that the amount of water used for the third rinsing cycle is about 20% of the amount of beer produced. The fraction of NaOH/water for the bath rinse to supply water is equal to 1/10. The whole cleaning system proceeds at atmospheric pressure. Details on the temperature set points and the energy losses in the different baths are given on Fig. 27.6.

The utility system includes the following subsections:

Hot utility system: The process heat is supplied conventionally by steam produced in a boiler. Hot water is also used in the process. It is stored in an intermediate hot water tank, which also serves as a heat recovery system.

Refrigeration system: Cooling requirements are supplied by distributing cold glycol–water mixture that is produced in an ammonia refrigeration cycle.

27.2.2 Time Average Approach

The beer production is described a priori as a succession of batch reactors. However, in industrial plants, although the processing operations are batch, the overall process can be considered as a nearly constant process by realising the operation in different batch reactions that are operated with a certain dephasing. The heat recovery being made is possible by introducing hot water buffer tanks. It is therefore possible to apply

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the time average approach (Linnhoff *et al.*, 1988) to represent the process operation by replacing the instantaneous flows or power by the mean energy needed per unit flow multiplied by the mean unit flow per product flow (Equation [27.1]).

$$\dot{m}_s\,[\mathrm{kg/s}] = m_s\,[\mathrm{kg/kg\,product}] \cdot \tilde{\dot{M}}_p\,[\mathrm{kg\,product/s}] \qquad [27.1]$$

where the mean flow of product flow is calculated by Equation [27.2]

$$\tilde{\dot{M}}_{\mathrm{product}}\left[\mathrm{kg_{product}/s}\right] = \frac{\dot{M}\left[\mathrm{kg_{product}/y}\right]}{t_p\,[\mathrm{s/y}]} \qquad [27.2]$$

In the following, all the flows and power that will be presented will refer to mean flows and power.

27.2.3 Inventory of Flows Entering and Leaving the System Boundaries

Following the definition of the system (Fig. 27.1), the first step is the identification of the flows entering and leaving. The flows entering the system are the raw materials (water, corn, malt (barley) and hops), the energy (natural gas in the boiler and electricity), the support materials (water, soda, etc.) and the packaging materials. The cleaning of bottles requires water and soda as input flows. The waste water leaves the cleaning system. The total water required at the third rinsing is 20% of the total beer produced, and is the same for the bath rinsing step as well. The amount of soda is 10% of the water added to the bath rinse. It leaves the system as part of the waste water flow. The streams leaving the system are the products (bottled beer) and the waste streams. The bio-waste of husk (dregs) accounts for 30% of the initial malt added. In addition, evaporated water and CO_2 from the fermentation leaves the system and is released into the environment.

Figure 27.7 summarises the major flows entering and leaving the system. Table 27.1 gives the specific major flows in units of bottled beer produced.

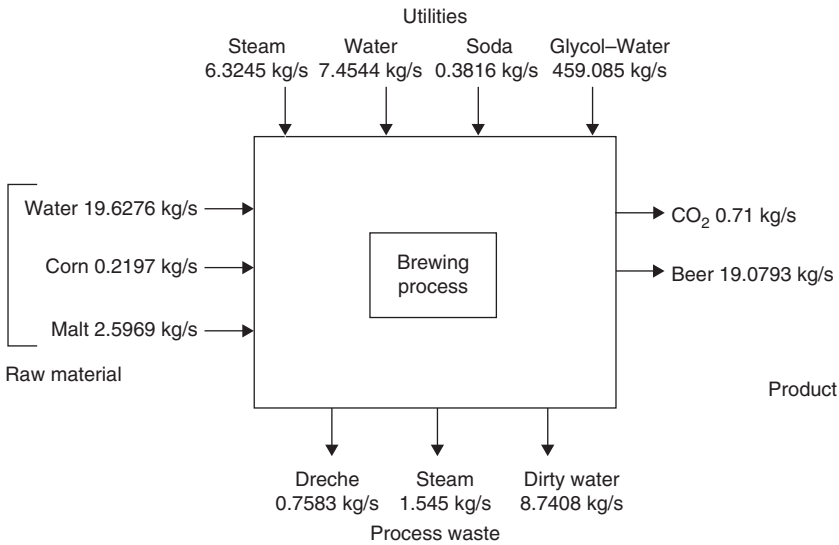
27.2.4 Energy Conversion and Distribution System

A boiler is used to produce the steam, thus natural gas is burned as primary fuel to generate the heat and convert the water into steam. Electricity is also required in small amounts for the pumping power for the water in the boiler unit. The waste stream from the boiler is the flue gases, which exhaust at 200°C in the chimney. The overall boiler efficiency is 89%, accounting for 2% of radiation losses and 9% in the flue gases.

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Table 27.1 Inlet and outlet flows of the process

Input		
Malt	0.135	kg/kg
Corn	0.0115	kg/kg
Water	1.029	kg/kg
Demineralised water	.051	kg/kg
Cleaning water	0.391	kg/kg
Soda	0.02	kg/kg
Natural gas	748.9	kJ_{LHV}/kg
Electricity	69.3	kJ e/kg
Cooling water	9.47	kg/kg
Output		
Beer	1.0	kg/kg
Brewing dregs	0.0397	kg/kg
Waste water	0.458	kg/kg
Vapour	0.081	kg/kg
CO_2	0.0373	kg/kg



27.7 Block flow diagram representation of the brewing process.

In most conventional bottle-cleaning systems, some steam is directly injected to heat up the water. This will require make-up water (demineralised water) to be added to compensate the losses.

Glycol–water is produced by a refrigeration unit, which consumes electricity as resource to drive its compressor. In nominal operation there is no requirement for make-up glycol–water unless there is leakage in the

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Table 27.2 Current energy bill

Description	Specific cost	Flows	Operating cost
Electricity	0.136 €/kWh	1,322 kWe	1,470.9 k€/y
Natural gas	0.048 €/kWh	17,062 kW_{LHV}	6,699.2 k€/y
Water	0.07 €/m3	0.31 m^3/s	630.0 k€/y
Demineralised water	1.0 €/m3	0.001 m^3/s	35.0 k€/y

Table 27.3 Current utility requirements with no heat recovery

Utilities	Requirements
Hot utility	17,062.0 kW_{LHV}
Cold utility	7,555.9 kW
Electricity	1,322.2 kWe

system. The refrigeration unit works with ammonia as the refrigerant fluid in our case.

Defining the Energy Bill

With no internal heat recovery, the conventional brewing process consumes 17,062 kW_{LHV} of primary energy in form of lower heating value of natural gas. This corresponds, therefore, to a heat supply to the process of 15,252.51 kW. The cooling load is of 7,555.9 kW, which is provided by the refrigeration unit, whereas 1,322.2 kW of electricity is required to drive the necessary pump and compressor. We find the natural gas consumption as 0.3595 kg/s, which results in a natural gas bill of 6,699.2 k€/y. The electricity cost, in this case, is 1,470.9 k€/y. The net yearly operating cost, including water and demineralised water, is 8,837.1 k€/y (see Table 27.2).

27.2.5 Defining the Process Heat-Transfer Requirements

The definition of the heat-transfer requirement of the process aims at analysis of the role of heat transfer in each of the process unit operations. The unit operations in the Flowsheet are therefore analysed with regard to their role in the beer-production recipe. From this analysis, one has then defined the heat-transfer requirement to be used for Process Integration, i.e. define the enthalpy–temperature profile of the heat-transfer demand including the corresponding ΔT_{min} contribution.

For defining the heat-transfer requirement, the main take-home messages are:

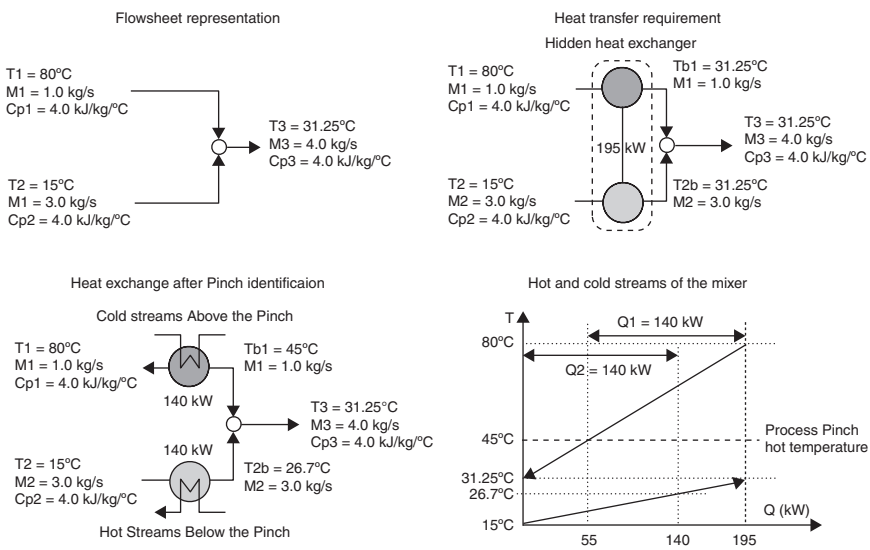
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Process unit analysis: analyse the role of the process unit operation and understand what are the inlet and output flows, and what are the principles of the unit operation, in order to identify the role of energy (i.e. heat and electricity) in the operation.

Hot stream definition: the goal is to cool down the hot streams with the highest possible temperature. This means looking at the process unit operation, and trying to define a process unit scheme that maximises the temperature of the fluids that will enter to be cooled in a heat exchange operation.

Cold stream definition: when heat is required, the goal is to define the minimum temperature at which the heat has to be supplied.

Non-isothermal mixing are heat exchangers: The non-isothermal mixing of Fig. 27.8, defines two heat-transfer requirements, one hot stream from 80°C to the mixing temperature, and one cold stream from 15°C to the mixing temperature (31.25°C). If the Pinch Point temperature is 40°C (i.e. 45°C for the hot stream), total amount of 140 kW over the 195 kW of the heat exchange in the mixing could be recovered for streams heating Above the Pinch while the same amount would be recovered from the hot streams Below the Pinch to heat up the corresponding cold stream in the mixer. In such a case, the mixer is a heat exchanger that transfers 140 kW across the Pinch.

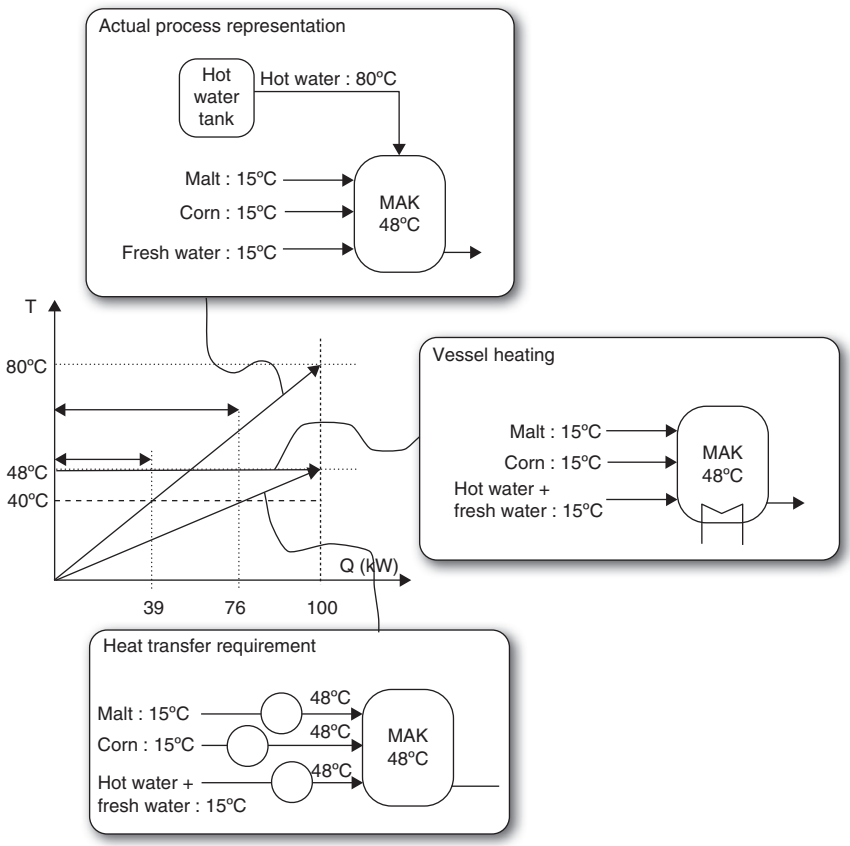


27.8 Non-isothermal mixing heat transfer requirement and heat recovery if the Pinch Temperature is at 45°C for the hot streams.

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It is therefore important, at the level of the targeting stage, to define all the non-isothermal mixing as heat exchangers and to decide, once the Pinch Points are identified, if non-isothermal mixing can still be realised or if part of the heat exchange has to be realised first in heat exchangers before mixing. In the given example, the final configuration would be that described in Fig. 27.8.

Let us analyse the process requirement of the first unit (MAK) on the Process Flowsheet. Referring to Fig. 27.9, there exist several ways of defining the heat-transfer requirements of the unit. First, it has to be observed that the heat requirement of the unit is in fact hidden in the hot water tank that supplies the MAK unit with hot water. The hot water tank is not really related to the process; it really serves as an intermediate stream for the



27.9 Process unit: heat-transfer requirement analysis, heat recovery if the Pinch Point is at 40°C.

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Table 27.4 Exergy demand of 100 kW heating for the different heat requirement of the same unit operation (MAK), possible heat recovery if the Pinch is at 40°C

Heat-transfer requirements	Exergy (kW)	Heat at 40°C (kW)
Hot water from 15°C to 80°C	11.40	38.46
Vessel heating at 48°C	11.84	0.00
Heat feed streams from 15°C to 48°C	6.97	75.76

process. The requirement of the MAK unit has therefore to be related to the injection of hot water, which defines the way the MAK requirement is satisfied today.

However, an alternative way of defining the requirement could be to decide to heat the vessel after mixing the ingredients, or better, to preheat the streams entering the vessels to a temperature of 48°C. In order to calculate the respective heat loads of the different streams, one has to first consider that the total amount of water is fixed; therefore, the water flow is the flow of the fresh water plus the flow of hot water. In this situation, the heat loads of the three heat-transfer requirements will be the same, but the defined cold streams will be different.

The solution with feed preheating is the one with the lowest exergy requirement (see Table 27.4), and also with the highest possible level for heat recovery. This is explained in Fig. 27.9 and illustrated where the Process Pinch Point is at 40°C. The heat recovery potential is in fact much higher when one defines the requirement taking into account the feed preheating. The feed preheating would, however, require a heat exchange between the corn, malt and water mixture and another process stream. Considering that this counter-current heat exchange could be difficult to realise, we have decided that only the water will be heated to a temperature, such that the mixture reaches the required 48°C.

The same kind of analysis is implemented for the different process unit operations.

27.2.6 Analysing the Streams Leaving the Systems

The streams leaving the system have to reach their system's boundary state. For the products, they have to be brought to their 'ideal' distribution state. The flows released to the environment have to reach a state that is in equilibrium with the environment. This can be done by heat exchange, and therefore defines a heat recovery opportunity. When setting up the heat recovery target, it is important to list and analyse all the heat exchange opportunities that are available from the streams leaving the system. In order to maximise the heat recovery, the enthalpy–temperature profile will be established following the rules presented above for the process streams, and defining as

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a target the conditions that are the closest to the environment conditions. The target state is however a *soft* target, since it does not necessarily have to be reached.

For the waste streams, it is important to examine if the waste treatment system is part of the system boundaries or not.

1. Beer – Beer leaves the system as the final product and will be stored to be filled later in the beer bottles.
2. Flue gases – Flue gases exhaust from the chimney of the boiler system (Steam Cycle) at almost 200°C. It is possible to treat these gases to cool down to optimal temperature to preheat the fresh air going to the burner. This in turn increases the exergy level of the hot gases in the burner and thus reducing the consumption of natural gas for the same amount of steam production.
3. Steam Condensate – Flash Vessel is used to improve the efficiency of condensation of the condensate return in the boiler system. This steam exhaust is usually lost and not recovered.
4. Husk (Dreche) – Malt waste in the form of husk is extracted during the filtration of mash before boiling process. The amount is approximately 30% by mass of the initially added barley. In our case it is around 0.7583 kg/s or 2.73 t/h. This husk is usually given to animals, losing the possibility of converting this into methane (biogas) through the biomethanation process.
5. Steam evaporation at WOK – 10% by mass of the working wort is evaporated during the boiling and energy is recovered from this vapour by cooling it down to the ambient temperature.
6. Trub – 0.2% by mass of wort is lost during the centrifuge step to finally remove any proteins from the wort.
7. Dirty water – A significant amount of dirty water leaves the bottle-cleaning system. This water includes soda, dirt and impurities absorbed during the process, and is usually sent to the water treatment plant to be used for other purposes.
8. Water – CIP – A part of water heated to 80°C in the hot section goes out of the system to be used for cleaning the tanks and other devices. This energy is not recoverable, and is lost in the cleaning necessary for maintaining the hygiene at the production line.

27.2.7 Overall Process Requirement Definition

The hot and cold streams defined for the whole process are given in Table 27.5. We give here some hints on the hot and cold streams definitions for the different sections.

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Table 27.5 List of streams in brewing process

Process Unit	Name – Process	$\dot{M}$(kg/s)	T_{in} (°C)	T_{target} (°C)	$\dot{Q}$(kW)	$\Delta T_{min}/2$
Hot Section 1	Water – MAK	2.4415	15	51.5	369.9	2
	MAK – Heating	2.9971	48	65	188.6	2
	Water – MAT	4.9652	15	54.5	823.6	2
	MAT – Heating	7.2982	48	65	420.1	2
Hot Section 2	MAT+MAK	10.2953	65	75	368.4	2
	Water – Filter	5.9130	15	80	1,608.9	2
	CIP	6.767	15	80	1,841.1	2
	WOK – Boiling	15.4500	78	105	1,662.9	2
	WOK – Evaporation	15.4500	105	105	3,319.8	2
	Vapour – Recovery	1.5450	105	100	17.2	1
	Vapour – Recovery	1.5450	100	100	3,487.9	
	Vapour – Recovery	1.5450	100	25	482.8	2
Hot Section 3	Centrifuge	1.3890	52	80	857.9	2
	Wort – Cooling	15.2634	102.5	10	5,434.6	2
Cold Section 1	Fermentation	15.2634	10	10	534.8	2
Cold Section 2	Cooling	15.2634	10	6	233.8	2
	Maturing	15.2634	6	6	425.5	2
Cold Section 3	Chilling	15.2634	6	1	714.5	2
	Chillage	15.2634	1	1	427.7	2
Cold Section 4	Beer preheating	15.26	1	15	822.05	2
	Pasteurisation heating	19.0793	15	80	4,854.05	2
	Pasteurisation cooling	19.0793	70	5	4,858.4	2
Cleaning	Soda Bath Temperature	3.1797	65	85	251.9	2
	Pre-rinse Bath Temperature	5.8272	40	60	472.8	2
	Soda make-up	0.3816	15	85	39.8	2
	Water to soda tank	3.816	15	85	1,287.1	2
	Pre-rinse make-up	0.1554	15	60	29.3	2
	Final Rinse Water make-up	3.816	35	60	399.34	2
	Soda Bath Rinse – Out	4.7696	85	60	472.7	2
	Pre-rinse – Out	8.7408	40	20	713.7	2

MAT and MAK tanks: the water is preheated to heat up the corn and malt streams and reach the required temperature in the different tanks. It is also assumed that the first preheating to 102°C is not necessary and that a temperature of 75°C has to be reached.

Evaporation in the launter tun: The evaporation takes place at 105°C. The vapour condensation is considered as a hot stream with a condensation at 100°C considering pure vapour. After condensation the liquid is still sub-cooled to reach the 25°C target temperature. This is a *soft* temperature.

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Centrifuge: A water make-up is added and has to be heated to 80°C.

CIP: Hot water used for Cleaning In Place is produced at 80°C. It is assumed that after its usage the hot water cannot be recovered and is sent cold to the water treatment plant.

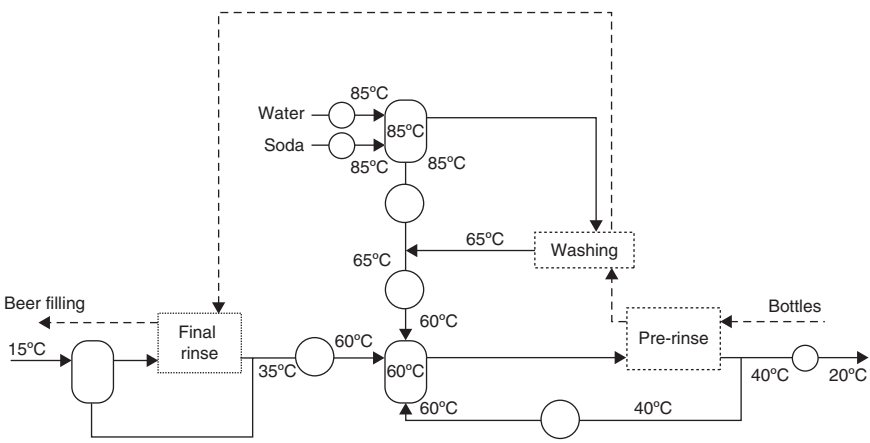
Trub: Trub could be added as a hot stream if heat exchange can be realised with the solid flow. The flow is however small when compared to the major flow.

Fermentation: The feed is first cooled down to the fermentation temperature, and then the temperature is maintained. Note that we consider the temperature in the tank, and not the temperature of the glycol–water mixture used to maintain the temperature in the tank.

Maturation: The flow is first cooled in a heat exchanger before reaching the maturation temperature.

Water dilution: This is a non-isothermal mixer. First, beer is preheated from 1°C to 15°C before being mixed with the diluting water. This preserves the cooling recovery capacity of the cold beer.

Bottle cleaning: The steam heating is replaced by the corresponding heat-transfer requirement. We consider a heat exchanger to control the bath tank temperature. The water make-up flow rate includes the water injected as steam in order to preserve the water balance. The make-up flows are preheated to the corresponding tank temperature. When water flows from one tank to another, isothermal mixing at the receiving tank is considered. This is done by introducing the corresponding heat exchanges. The hot water leaving the bottle-cleaning system and going to the treatment plant is cooled down to 20°C. The heat exchange requirement in the bottle-cleaning system is represented on Fig. 27.10.



27.10 Heat-transfer requirement of the bottle-cleaning section.

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27.3 Calculating Maximum Heat Recovery in the System

Assuming a ΔT_{min} contribution of 2°C for each liquid stream (i.e. an overall ΔT_{min} of 4°C), we calculate the maximum heat recovery between the hot and cold streams of the process. We also plot the Composite Curves and Grand Composite Curve of the process (Figs 27.11 and 27.12). We obtain by balance the minimum energy requirements (MER) for the brewing process, considering the heating, cooling and refrigeration requirements (Table 27.6). For the refrigeration requirement, we consider the fresh water as defining the cooling water temperature. From our analysis, the Pinch Point is at the ambient temperature. This means that the overall cooling requirement is fact to be realised below the ambient temperature, and has to be realised using a refrigeration cycle.

The current energy requirements are found to be 15,185.18 kW_{LHV} for the heating load and 7,555.9 kW for the refrigeration load to be compared with the minimum target values of 4,904.0 and 3,089.1 kW of refrigeration load. The energy savings correspond therefore to 67.71% of savings for the hot utility and 57.57% of savings for the refrigeration load.

27.3.1 Identification of the Penalising Heat Exchangers and the Heat Recovery Potentials

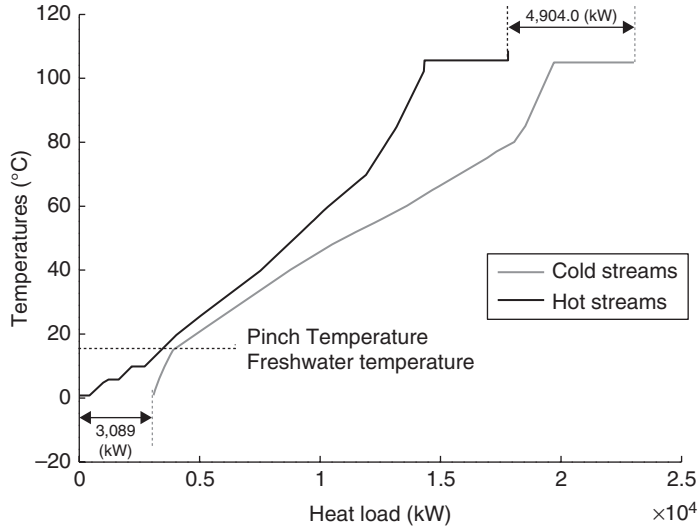
The penalising heat exchangers are heat exchangers that transfer heat across the Pinch Point. The definition of the penalising heat exchangers has however to be completed by the existence of Pseudo-Pinch Point as the one created by the heat recovery of the evaporated vapour.

One major penalty is created by the refrigeration system that cools down the beer product after pasteurisation. This is a heat exchanger crossing the Pinch Point. The heat recovery penalty is 3,957.9 kW, i.e. 38.50%. This saving is realised both Above the Pinch i.e. reduction of the heating requirement, and Below the Pinch, i.e. reduction of the refrigeration requirement where it accounts for 88.61% of the savings.

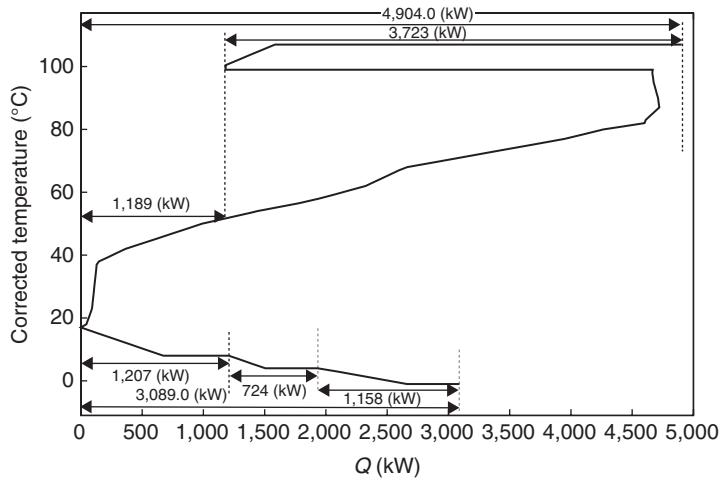
Table 27.6 Minimum heating and cooling requirement target for the process

Minimum requirement	Amount (kW)
Heating	4,904.0
Cooling	0
Refrigeration	3,089.1

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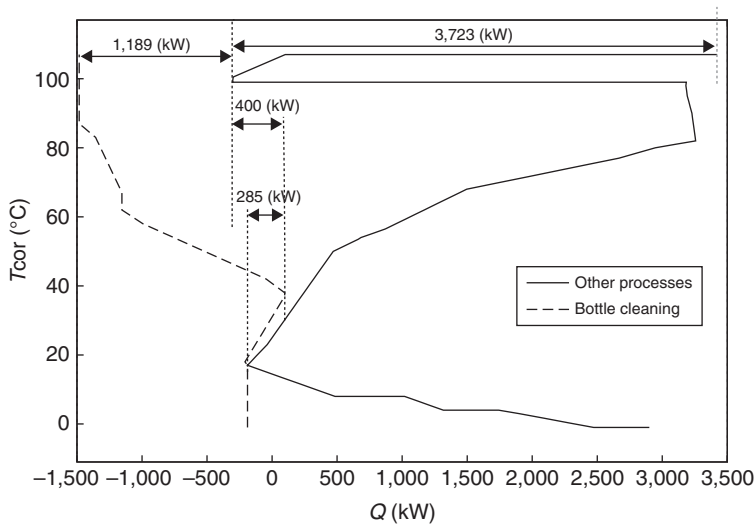
27.11 Hot and Cold Composite Curves of the process.



27.12 Grand Composite Curve of the brewing process.

The additional saving comes from heat recovery from evaporation and the bottle-cleaning system. Considering that the Pinch Point is created by the freshwater input temperature, the overall amount of heat available could be recovered in the process, leading to an additional saving of 4,701.6 kW explaining 45.7% of the energy penalty.

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27.13 Integrated Composite Curve of the bottle-cleaning section.

It is also interesting to analyse the integration between sections. For this analysis, the Integrated Composite Curve representation is interesting. As an example, Fig. 27.13 shows the Integrated Composite Curve of the bottle-cleaning section. One can see from this analysis that the minimum heat requirement can be divided into two major parts that could be solved dependently one from the other: the process hot section that receives 3,723 kW of the high temperature heat, and 1,189 kW sent to the bottle-cleaning subsystem. 285 kW will be recovered from the waste heat of the bottle-cleaning system to preheat the streams of the other sections, while the 400 kW will be supplied from the higher temperature waste heat from the hot section of the bottle-cleaning section.

From analysis of the Grand Composite Curve and the Integrated Composite Curve, it is also important to notice that most of the steam used in heat exchangers or steam injection to supply heat to the process at a temperature below 96°C will have probably been changed into heat recovery heat exchangers.

27.4 Defining the Energy Conversion System

Before designing the heat recovery Heat Exchanger Network (HEN) system, it is important to first analyse the integration of the energy conversion system.

Conventionally, a boiler is used to provide steam for the heating requirements, and a glycol–water system powered by a refrigeration unit is used to provide the refrigeration requirements. It is, however, important to analyse

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also the possibility of integrating heat pumps and cogeneration units in order to reduce the energy bill of the process.

27.4.1 Grand Composite Analysis

The analysis of the Grand Composite Curve (Fig. 27.12) already gives some indications of the possible energy conversion opportunities.

For temperature Above the Pinch, the heat requirement can be divided into two different zones:

- Above 105°C: 3,723 kW has to be supplied to the process, mainly for the beer evaporation above 105°C.
- From 20°C to 60°C: 1,189 kW can be supplied, mainly for water preheating. This could indicate an opportunity for working with a lower temperature in the boiler chimney, realising flue gas water condensation for preheating process water. From this analysis it can also be concluded that air preheating would not be a valid option for the recovery boiler. Introducing condensation in the boiler and considering cooling down the gas to 40°C could increase the boiler efficiency to 104% based on the lower heating value of the natural gas, leading therefore to a consumption of 4.715.4 kW_{LHV} or a gas bill reduction of 72.4%.

Below the Pinch, the overall amount of refrigeration has to be supplied by a single refrigeration cycle. In the present situation, the refrigeration cycle is working with a Coefficient of Performance (COP) of 5.80 kW_{cold}/kWe. This leads to a cost of 0.023 €/kW h_{cold}. Assuming that the efficiency of the refrigeration cycle will not be affected by the reduction of the demand, the expected electricity consumption for the refrigeration cycle can be estimated as 532.42 kW.

From the analysis of the Grand Composite Curve, the refrigeration demand can be divided into three parts:

- From ambient (19°C) to fermentation (10°C): This corresponds to 1,207 kW or 39.07% of the demand.
- From fermentation (10°C) to maturation (6°C): This corresponds to 724 kW or 23.44% of the demand.
- From maturation (6°C) to the lowest temperature (1°C): Corresponds to the rest of the supply, i.e. 1,158 kW (37.49%).

Staging the supply of the refrigeration would allow increasing the efficiency of the refrigeration system, and therefore reducing the price of supplying

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the refrigeration load. It is possible to approximate the impact of staging the refrigeration load by using the concept of the Carnot Factor. For a refrigeration load $\dot{Q}_c$ at a temperature T_{cold} supplied to a source with a temperature T_{amb}, the theoretical mechanical power $\dot{E}_{ref}$ required is calculated by Equation [27.3]. Knowing the present consumption, one could calculate an efficiency η by Equation [27.4] and consider that this efficiency will stay constant. The new consumption can therefore be calculated by Equation [27.5]. Table 27.7 gives the estimated mechanical power for the three levels identified. In the calculation, we have considered a temperature difference of 9°C between the refrigeration requirement and the supply considering the possible use of an intermediate fluid.

$$\dot{E}_c(T_c) = -\dot{Q}(T_c)\cdot\left(1-\frac{T_{amb}}{T_c}\right) \quad [27.3]$$

$$\eta = \frac{\dot{E}_c(T_{ref})}{\dot{W}_c(T_{ref})} \quad [27.4]$$

$$\dot{W}_c(T_c) = \frac{-\dot{Q}(T_c)\cdot\left(1-\left(T_{amb}/T_c\right)\right)}{\eta} \quad [27.5]$$

Using the approximation of the exergy efficiency, one could approximate the mechanical power required to 449.8 kW. The staging corresponds to an expected refrigeration efficiency increase of 15.51% and a new COP of 6.9 kW_{cold}/kW e.

This would also reduce accordingly the use of cooling water by 0.12 m^3/s, which corresponds to a reduction of 0.60%.

With this first analysis, it is therefore possible to estimate the new energy bill after heat recovery and integration with the present energy conversion system, see Table 27.8.

It could be, however, assumed at this stage that the heat of the refrigeration cycle is directly sent to the cooling water and not reused in the process.

Table 27.7 Expected mechanical power for the refrigeration loads with a reference temperature of 25°C (298 K)

Temperature (°C)	Load (kW)	$\dot{W}(T = -8°C)$	$\dot{W}(T)$
10	1,207.0	208.0	146.3
6	724.0	124.8	103.9
1	1,158.0	199.6	199.6
Total	3,089.0	532.4	449.8

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Table 27.8 Estimated new energy bill after heat recovery

	Annual bill (k€/y)	%ref
Natural gas	1,851.45	28
Electricity	500.44	34
Water	251.12	40
Total	2,603.01	29

27.4.2 Defining the Possible Energy Conversion Technologies

Existing Energy Conversion Systems

Steam boiler: A boiler is used to produce steam and thus to provide the heat requirements of the brewing process. A detailed diagram is given in Fig. 27.14. Steam is produced at a temperature of 193°C and a pressure of 8.5 bar; it is then expended to 6 bar in the distribution system. The condensates return at 3 bars in saturated conditions. It is then flashed to a pressure of 1.5 bar in a deaeration tank, where it is mixed with the demineralised water make-up required to compensate the steam used in direct injections in the process. A pump (efficiency 90%) is necessary to raise the pressure of the returned condensate before it enters into the burner for heat transfer.

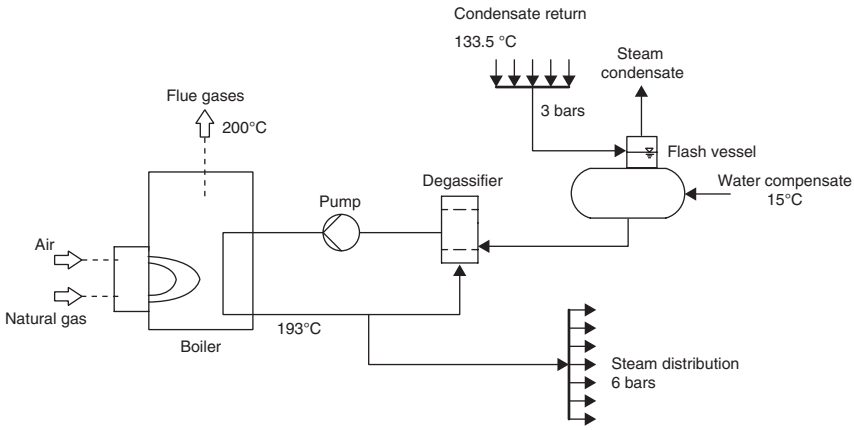
Refrigeration cycle: The refrigeration needs are supplied by an ammonia refrigeration cycle (Fig. 27.15) operating between 3.2 bar (–8°C) and 10 bar (25°C). The compressor isentropic efficiency is 85%. The cycle is cooled using cooling water and supplies a glycol–water distribution loop operating between –5°C and –1°C.

Alternative Energy Conversion Units

Cogeneration engine: A cogeneration engine could be used to replace the boiler system. Internal combustion engines have an attractive electrical efficiency, and it is possible to recover heat from both exhaust gases and cooling water. Table 27.9 shows the characteristics of a typical cogeneration engine using natural gas as fuel.

According to the optimal placement of a cogeneration unit, the hot streams of the cogeneration unit should be Above the Pinch Point. From the analysis of the Grand Composite Curve, however, one may expect problems in integrating the cogeneration unit since the temperature of the water cooling system is not high enough to supply the evaporation heat load in

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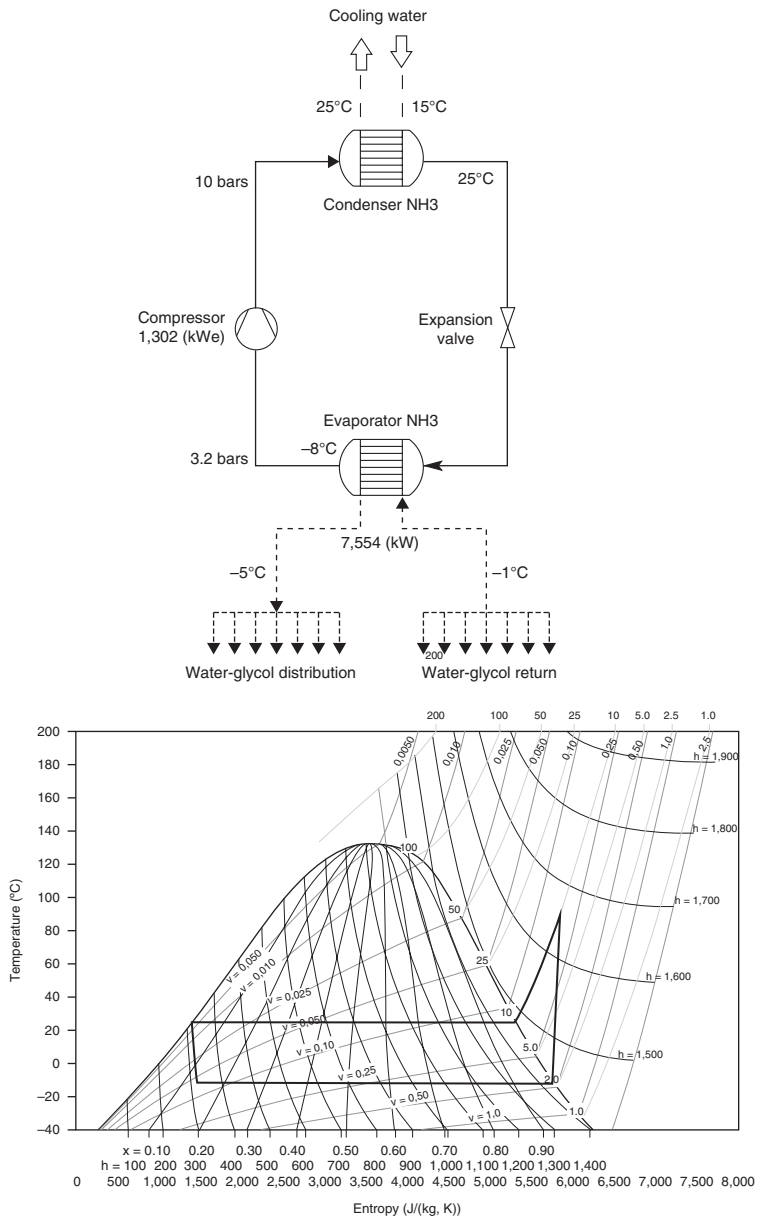
27.14 Steam distribution system of the process.

the process. However, based on the Grand Composite Curve analysis, it could also be suggested to use water condensation in the recovery boiler in order to supply the low temperature heat.

Waste treatment and energy conversion: The husk leaving the process is hot and is rich in carbon and hydrogen species. The husk has a lower heating value of approximately 14.3 MJ/kg and that can be converted into a mixture of CH_4 and CO_2 by bio-methanation in anaerobic conditions. This reaction generally requires a specific operating temperature of 35°C, which defines a new cold stream in the process.

Based on the report published by the French agency ADEME (ADEME, 2011), 75 Nm^3 of bio-methane with a lower heating value of 35.7 MJ/Nm^3 can be recovered from 1 t of husk, which represents 22,430 t of husk for this brewery study, resulting in 1,674,781 m^3 of possible methane production. With proper calculations, by using the lower heating value of methane, we may produce 16,608 MWh_{LHV}/y or an available mean power of 2,030.3 kW. This production requires about 4% (81.2 kWe) of lower heating value as electricity for the bio-methane production, as well as 20% (406.1 kW) as heat to maintain the temperature in the methanation vessel. This defines a cold stream to be added in the list of the streams to be heated. The heat available in the husk is at 75°C, to be cooled down to 35°C, and this makes a heat load of 54.8 kW (i.e. $0.762 \times 1.8 \times (75 - 35)$). If the bio-methane produced is used in a cogeneration unit for Combined Heat and Power production, we may expect an electrical efficiency of 34% and a thermal efficiency of 52%. Table 27.10 defines the resulting hot and cold streams of the bio-methane production and its combustion in a gas engine.

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27.15 Ammonia cycle for producing the glycol–water cold loop.

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Table 27.9 Cogeneration unit characteristics based on GE-Jenbacher data, power range: 500–1100 kW (www.gejenbacher.com)

Description				Power (kW)	Efficiency (%)
$\dot{M}_{f\,uel}$ (kWe LHV)				2,605	
$\dot{W}_e$ (kWe)				1,063	40.8
Without water condensation in the flue gases					
	T_{in} (°C)	T_{out} (°C)	$\Delta T_{min}/2$	Power (kW)	Efficiency (%)
$\dot{Q}_{Gas}$ (kW)	470	120	15	537	21.0
$\dot{Q}_{Cooling}$ (kW)	87	80	2	653	25.0
$\dot{Q}_{th}$ (kW)				1,190	46.0
With water condensation in the flue gases					
	T_{in} (°C)	T_{out} (°C)	$\Delta T_{min}/2$	Power (kW)	Efficiency (%)
$\dot{Q}_{Cooling}$ (kW)	87	80	5	653	25.07
$\dot{Q}_{Gas}$(kW)	470	60	15	632	24.26
$\dot{Q}_{Gas}$(kW)	60	40	15	209	8.02
$\dot{Q}_{th}$ (kW)				1,494	57.35

Table 27.10 Cogeneration unit based on the bio-methane produced on the process husk

Description				Power (kW)	Efficiency (%)
$\dot{M}_{f\,uel}$ (kWe_{LHV})				2,030.3	
$\dot{W}_e$ (kWe)				609.1	30.0
	T_{in} (°C)	T_{out} (°C)	$\Delta T_{min}/2$	Power (kW)	Efficiency (%)
$\dot{Q}_{Cooling}$ (kW)	87	80	5	424.1	20.89
$\dot{Q}_{Gas}$ (kW)	470	90	15	631.7	31.11
$\dot{Q}_{Gas}$ (kW)	34	35	2	351.2	17.30

27.4.3 Heat Pumping

Refrigeration Cycles Integration

As the Pinch Point is also at the ambient temperature, the hot stream of the refrigeration cycle can also be used to supply heat to the process. The operating conditions of the refrigeration cycle have therefore to by adjusted, not only for the evaporation temperature but also for the condensation temperature. Therefore, we have to optimise the operating conditions of the refrigeration cycle. As the refrigeration system is using a glycol–water mixture to distribute the cold to the process, the solution will be to reuse the existing system and to reorganise the glycol–water distribution system with three different temperature levels.

The operating conditions will change the amount of heat that can be recovered, and therefore affect the flow of the other refrigeration cycles to be used.

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Table 27.11 New operating conditions for the three refrigeration cycles

	Evaporation (°C)	Condensation (°C)	$\dot{Q}_c$ (kW)	$\dot{W}_{mec}$ (kW)
Conditions 1	1.0	50.0	1,000.0	355.1
Conditions 2	–3.0	38.3	1,000.0	312.6
Conditions 3	–8.0	26.7	1,000.0	276.6

The final operating conditions considered for the refrigeration cycle are given in Table 27.11. It would be still possible to optimise the operating conditions of the cycles if needed.

Mechanical Vapour Recompression

The theory of the heat pump integration recommends the integration of the heat pump to raise heat from Below to Above the Pinch Point. If, from the Grand Composite Curve analysis, the mechanical vapour recompression appears not to be interesting, it will be shown that the mechanical vapour recompression is justified when the energy conversion system integration is considered. In fact, the heat pumping effect of the refrigeration cycle will introduce utility Pinch Points that will justify the use of the mechanical vapour recompression. The mechanical vapour recompression will also help the integration of the cogeneration engine, since the heat of the engine cooling water is available below the high temperature Pinch Point.

The mechanical vapour recompression system is described in Fig. 27.16. (See also Table 27.12.) It should be noted that only part of the recovered vapour will be compressed, the rest of the flow being recovered is without recompression since it has already reached the required temperature level. It should be mentioned that the representation of the mechanical vapour recompression is identical to a close cycle. This means that one can always visualise the evaporation heat load and the exchange with the condensing vapour. In reality, there will be no heat exchanger at the evaporation temperature, the evaporation being directly replaced by a reduction of the condensation heat load.

It also has to be highlighted that, for the two streams (recompressed and not), the condensed water is cooled down to the system limits.

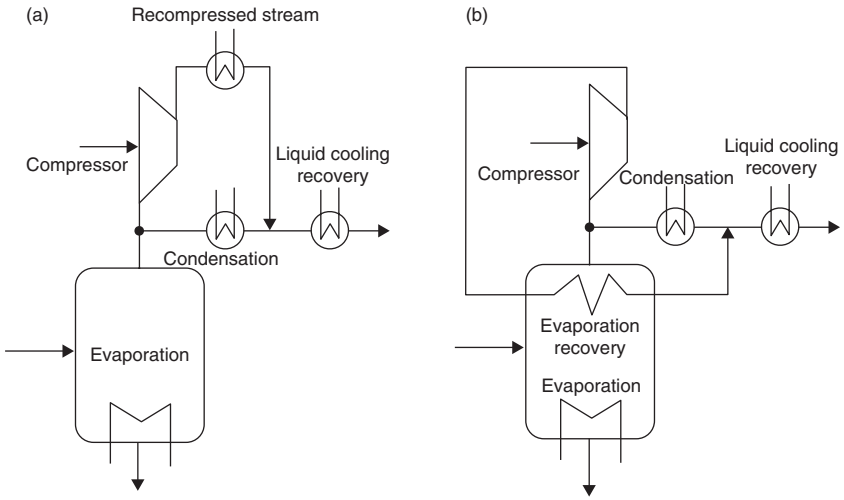
27.4.4 Mathematical Formulations

From the analysis of the Grand Composite Curve and the proposed set of energy conversion units, we can conclude that there will be more than one option for supplying the requirement of the process. In addition, one can

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Table 27.12 Operating conditions for the mechanical vapour recompression

	Evaporation	Condensation	Max$\dot{Q}_c$	Max$\dot{W}_{mec}$	COP$^{\cdot}$
	(°C)	(°C)	(kW)	(kW)	
MVR 1	98.0	109.0	3,434.2	150.9	23.8



27.16 Mechanical vapour recompression scheme for vapour recovery: (a) scheme for external use of the recompressed stream and (b) scheme for internal use of the recompressed stream.

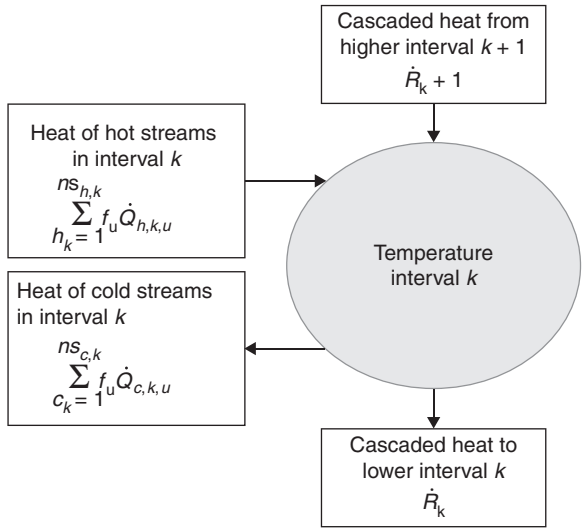
also see that the different energy conversion systems will be interconnected or interrelated.

When looking at the energy conversion units, one can see that each is related to hot and cold streams, as well as to a cost or resource consumption and to mechanical power consumption or production. At the difference with the process streams, the flows in the utility systems have to be optimised in order to minimise the cost of supplying the energy requirement.

Following the theory of Process Integration, this means maximising the flow of the cheapest utility first. If this works for simple systems, this is not anymore the case for the system under study, since both hot and cold streams of the refrigeration cycles are used and the Combined Heat and Power production units will interact and/or compete with the heat pumps and the mechanical vapour recompression.

It is therefore easier to calculate the flows by using mathematical programming techniques. The Mixed Integer Linear Programming (MILP) formulation proposed by Marechal and Kalitventzeff (1998b) solves the

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27.17 Graphical Heat Cascade representation for MILP formulation without heat exchange restrictions.

Heat Cascade and calculates the maximum heat recovery, together with the optimal integration of energy conversion units, by minimising the operating cost (Equation [27.6]) of supplying the heat and cold requirements of the process while satisfying the Heat Cascade constraints (Equation [27.7]), as illustrated in Fig. 27.17. Introducing the electricity balances for import (Equation [27.9]) and export (Equation [27.9]), the CHP production is, therefore, optimised considering the possibility or not of exporting electricity to the grid. In the formulation, the bold text corresponds to variables of the problem to be defined by the optimisation algorithm, while normal text refers to problem parameters that will not be affected by the optimisation procedure.

When temperatures are fixed, the problem is a linear programming problem, since the objective function and the constraints are all linear with respect to the problem variables. The use of integer variables allows engineers to select the best utility system and to propose more than only the best selection.

Solving the MILP problem will calculate the best flows in the system, considering the possible interactions between the process streams and the utility system, but also considering the heat exchange interactions between two utility flows.

As this is a linear problem, the optimal solution is identified by the activation of inequality constraints. In our case, two major categories can be activated:

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The Pinch Points correspond to the activation of $\dot{R}_k \geq 0$ constraints and will define Pinch Points between the Hot and Cold Composite Curves.

The maximum flow corresponds to the maximum usage allowed for a utility stream.

It has to be highlighted that, due to the Heat Cascade constraints (Equation [27.8]) it is assumed that the proposed utility system is chosen with the appropriate conditions to satisfy the process requirement. As a consequence no 'magic' heating or cooling duty will be added to the system and the resulting Hot and Cold Composite Curves will be balanced.

$$\min_{\dot{E}_{\overline{el}}, \dot{E}^{+}_{el}, f_u, \dot{R}_k, f_u} F_{obj} = \sum_{f=1}^{nf} \left(c^{+}_{f} \sum_{u=1}^{nu} f_u \dot{E}^{+}_{f,u} \right) + c^{+}_{el} \dot{E}^{+}_{el} - c^{-}_{el} \dot{E}^{-}_{el} + \sum_{u=1}^{nu} f_u c_u \qquad [27.6]$$

$$\sum_{h_k=1}^{ns_{h,k}} f_u \dot{Q}_{h,k,u} - \sum_{c_k=1}^{ns_{c,k}} f_u \dot{Q}_{c,k,u} + \dot{\boldsymbol{R}}_{\boldsymbol{k+1}} - \dot{\boldsymbol{R}}_{\boldsymbol{k}} = 0 \quad \forall k = 1,\ldots,nk \qquad [27.7]$$

$$\dot{\boldsymbol{R}}_{\mathbf{1}} = 0 \;\; \dot{\boldsymbol{R}}_{\boldsymbol{nk+1}} = 0 \quad \dot{\boldsymbol{R}}_{\boldsymbol{k}} \geq 0 \;\; \forall k = 2,\ldots,nk \qquad [27.8]$$

$$\dot{E}_{el,process} + \sum_{u=1}^{nu} f_u \dot{E}^{+}_{el,u} + \dot{\boldsymbol{E}}^{+}_{el} - \sum_{u=1}^{nu} f_u \dot{E}^{-}_{el,u} \geq 0 \qquad [27.9]$$

$$\dot{E}_{el,process} + \sum_{u=1}^{nu} f_u \dot{E}^{+}_{el,u} + \dot{\boldsymbol{E}}^{+}_{el} - \dot{E}^{-}_{el} - \sum_{u=1}^{nu} f_u \dot{E}^{-}_{el,u} = 0 \qquad [27.10]$$

$$\dot{E}^{+}_{el} \geq 0 \quad \dot{E}^{-}_{el} \geq 0 \qquad [27.11]$$

$$y_u \cdot f_u^{min} \leq f_u \leq y_u \cdot f_u^{max} \qquad [27.12]$$

Based on a formulation originally presented by Papoulias *et al.* (1983), the MILP formulation presented here can be easily extended to introduce additional constraints, such as restricted matches (Becker and Maréchal, 2012a) or multi-period problems (Marechal and Kalitventzeff, 2003). It can also be used to solve combined heat and water problems (Marechal and Kalitventzeff, 2005) as well as to optimise the operating conditions of complex systems, such as integrated steam networks (Marechal and Kalitventzeff, 1999) or site scale problems (Marechal and Kalitventzeff, 1998a).

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Comparison of Energy Conversion System

In this section we compare different options for the supply of heat to the process. The following scenarios for integration of energy conversion units are considered.

Scenario A Boiler + Refrigeration
Scenario B Boiler + Heat pumping from the refrigeration cycle
Scenario C Condensation boiler + Refrigeration
Scenario D Condensation boiler + Heat pumping from refrigeration cycle
Scenario E Condensation boiler + Heat pumping from refrigeration cycle and mechanical vapour recompression
Scenario F Cogeneration engine + Heat pumping from the refrigeration cycle
Scenario G Cogeneration biogas engine and natural gas + Heat pumping from the refrigeration cycle
Scenario H Cogeneration biogas engine and natural gas + Heat pumping from the refrigeration cycle and mechanical vapour recompression

Table 27.13 presents the results of the energy conversion system integration including the heat recovery options identified by the Process Integration analysis.

Table 27.14 compares the energy consumption results of the different scenarios.

Primary energy is calculated by considering an efficiency of 56% for the electricity production (this is the efficiency of a natural gas combined cycle) and accounting only for fossil fuel (i.e. the biogas produced from the waste is not accounted as a primary energy resource). In this table, the scenario S0 refers to the present situation without heat recovery.

Table 27.13 Detailed results (power of the energy conversion system in kW)

Case	frg1	frg2	frg3	MVR	Cog	Bcog	combcond	comb
SA	215.7	153.8	173.0	0.0	–0.0	–0.0	0.0	4,726.1
SB	368.1	224.0	208.4	0.0	–0.0	–0.0	0.0	3,804.4
SC	215.7	153.8	173.0	0.0	–0.0	–0.0	4,234.9	0.0
SD	346.4	211.9	202.3	0.0	–0.0	–0.0	3,784.4	0.0
SE	531.6	289.6	237.7	69.0	–0.0	–0.0	1,676.8	0.0
SF	531.6	289.6	237.7	0.0	6,272.1	-0.0	0.0	0.0
SG	531.6	289.6	237.7	0.0	5,120.3	609.1	0.0	0.0
SH	480.0	244.1	312.5	107.1	1,353.4	–0.0	0.0	0.0
SI	486.5	265.9	289.6	101.3	455.9	609.1	0.0	0.0

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Table 27.14 Energy consumption of solutions

Case	Nat.gas (kW)	Cool. water (kW)	Elec. out (kWe)	Elec. in (kWe)	Primary energy (kW)	% (%)
S0	17,062.0	0.0	8,878.0	1,322.2	19,423.0	100.0
SA	4,726.1	3,141.6	0.0	579.7	5,761.3	29.7
SB	3,804.4	2,539.0	0.0	834.2	5,294.2	27.3
SC	4,234.9	3,141.6	0.0	579.7	5,270.1	27.1
SD	3,784.4	2,890.3	0.0	797.4	5,208.5	26.8
SE	1,676.8	1,061.2	0.0	1,150.0	3,730.3	19.2
SF	15,680.1	8,063.3	5,058.3	0.0	6,647.4	34.2
SG	12,800.8	7,148.4	4,530.3	0.0	4,710.9	24.3
SH	3,383.4	1,232.3	172.2	0.0	3,076.0	15.8
SI	1,139.7	674.6	0.0	107.3	1,331.3	6.9

Table 27.15 Total cost calculation including investment

Case	Nat.Gas (k€/y)	Elec. (k€/y)	Opex (k€/y)	Maint. (k€/y)	Inv. (k€)	Total (k€/y)	% (%)
S0	6,699.2	1,405.8	8,105.0	0.0	0.0	8,105.0	100.0
SA	1,855.7	616.5	2,472.1	0.0	0.0	2,472.1	30.5
SB	1,493.8	887.1	2,380.9	0.0	0.0	2,380.9	29.4
SC	1,662.8	616.5	2,279.3	0.0	0.0	2,279.3	28.1
SD	1,485.9	848.0	2,333.9	0.0	0.0	2,333.9	28.8
SE	658.4	1,222.9	1,881.3	1.0	75.9	1,889.1	23.3
SF	6,156.7	–5,379.0	777.6	94.1	6,899.3	1,488.0	18.4
SG	5,026.1	–4,817.5	208.5	85.9	6,302.3	857.4	10.6
SH	1,328.5	–183.1	1,145.4	21.9	1,606.5	1,310.8	16.2
SI	447.5	114.1	561.6	17.5	1,283.0	693.7	8.6

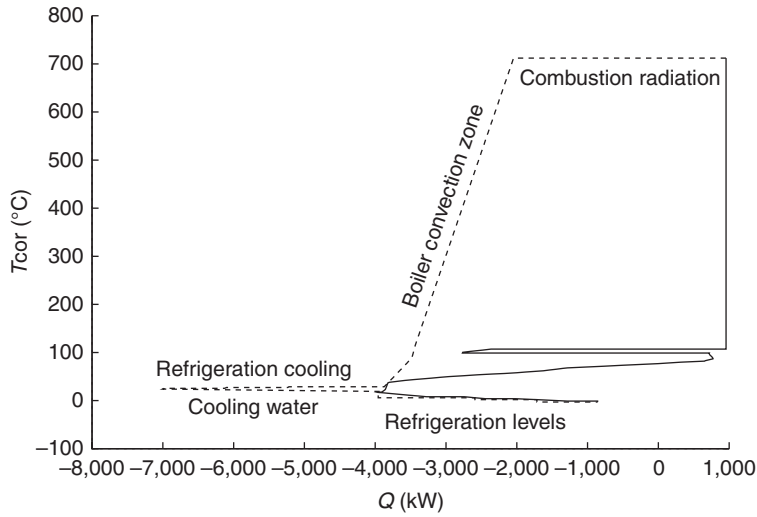
Table 27.15 presents the overall results obtained for the different scenarios. For electricity, only the electricity needed by the energy conversion system is considered, not the electricity needed by the process itself.

For calculating the operating cost, the yearly operation time is considered and it does not distinguish between selling or purchasing electricity. The values of Table 27.16 are considered to estimate the investment. Only cogeneration and mechanical vapour recompression are considered for the investment, and calculations have not been considered for the other units or for the investment of the heat recovery heat exchangers. It could be mentioned that the reuse of the existing refrigeration system should be analysed in more detail to confirm that the new compression ratio can be realised with the existing system. The operation with different operating temperature in the refrigeration implies the reorganisation of the glycol–water system adding more tanks and defining a new operating strategy. It also has to be mentioned that staging the heat recovery from the refrigeration cycle,

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Table 27.16 Data for the investment estimation

Cogeneration	1.100	(k€/kWe)
Cogeneration maintenance	0.015	(€/kWhe)
Mechanical vapour recompression	$4.000*(\dot{E}_{el})^{0.9}$	(k€)
Mechanical vapour recompression maintenance	0.02	(€/kWhe)
Estimated lifetime of equipment	15	(y)
Interest rate	6	(%)



27.18 Integrated Composite Curves of energy conversion system for scenario A.

profiting from the heat pumping effect, will mean retrofitting the hot water recovery system, introducing one hot water storage tank per Utility Pinch Point introduced in the system. This requires the development of a more detailed simulation model.

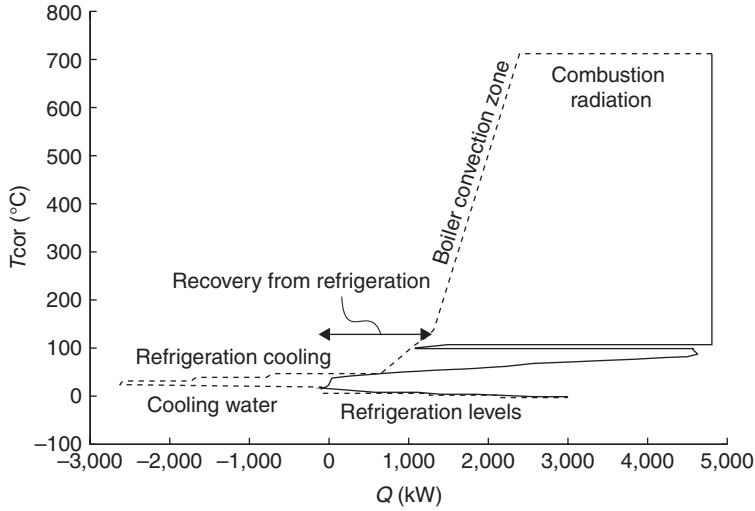
Given below some comments on the different scenarios.

Scenarios A and B – Boiler + Refrigeration

These two scenarios allow showing the benefit of the heat recovery from the refrigeration cycle. By increasing the temperature level of the refrigeration cycle, one can see that 922.0 $kW_{LH\,V}$ can be saved with the heat pumping solution. The additional mechanical power consumed is 254.5 kWe, which features an overall COP of 3.6.

The Integrated Composite Curves of the energy conversion system for these two solutions are compared on Figs 27.18 and 27.19.

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27.19 Integrated Composite Curves of energy conversion system for scenario B.

Scenarios C and D – Condensing Boiler + Refrigeration

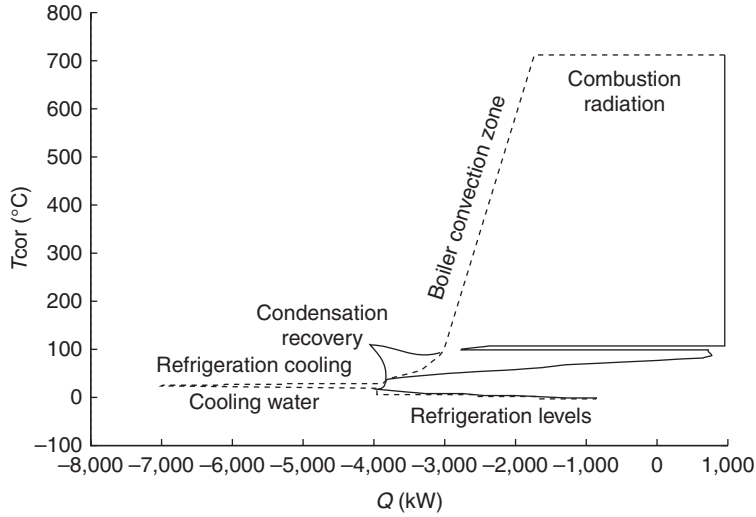
Introducing a condensing boiler could save 491.2 kW or 10.39% of the natural gas consumption in the minimum energy requirement conditions. By increasing the temperature level of the refrigeration cycle, one can see that 451.0 kW_{LHV} can be saved by the heat pumping solution. The additional mechanical power consumed is 217.7 kWe, which features an overall COP of 2.1. It is interesting to note that the saving from the condensing boiler is no longer significant, and that most of the condensing heat is finally released in the cooling system.

The analysis shows, therefore, that although a real benefit can be observed from the condensing boiler integration, the benefit is set to zero when the heat pumping from the refrigeration system is considered. The difference between scenario B and scenario C is explained by a small reduction of the heat pumping temperature in the refrigeration cycle, which leads to an increase in the efficiency of the refrigeration cycle. The Integrated Composite Curves of the energy conversion system for these two solutions are compared on Figs 27.20 and 27.21.

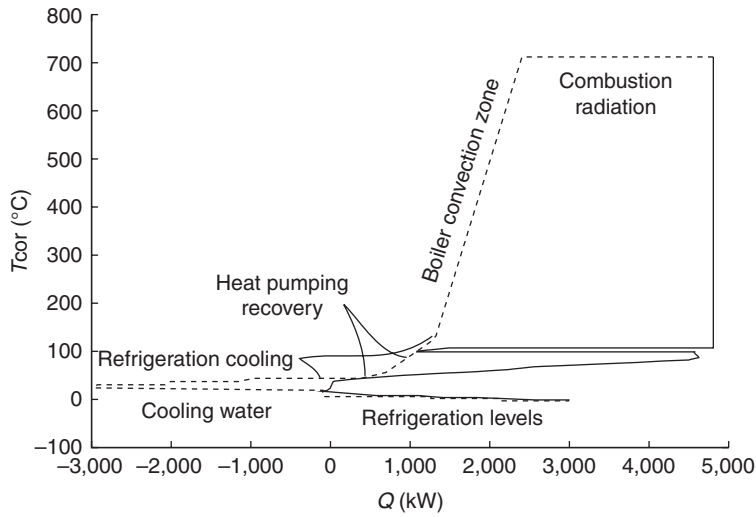
Scenario E – Condensing Boiler, Refrigeration and Mechanical Vapour Recompression

When mechanical vapour recompression is introduced, the Utility Pinch Point observed in scenarios B and D is moved in order to reduce the flow of natural gas. The use of mechanical vapour recompression is optimised,

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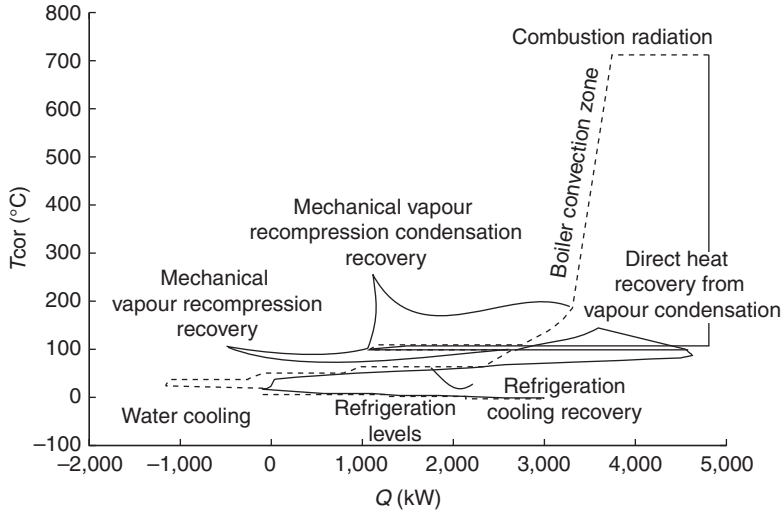
27.20 Integrated Composite Curves of energy conversion system for scenario C.



27.21 Integrated Composite Curves of energy conversion system for scenario D.

together with the temperature of the refrigeration cycle heat recovery, to profit from the heat pumping effect (Fig. 27.22). The natural gas saving corresponds to 2,559.0 kW or 60.42% of the natural gas consumption in the minimum energy requirement conditions. This is obtained with an increase

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27.22 Integrated Composite Curves of energy conversion system for scenario E.

of 570.3 kWe spent in the refrigeration cycles and in the mechanical vapour recompression. The COP of this scenario is of 4.49 when compared with the solution using the condensing boiler with heat pumping.

Scenarios F and G – Cogeneration and Refrigeration

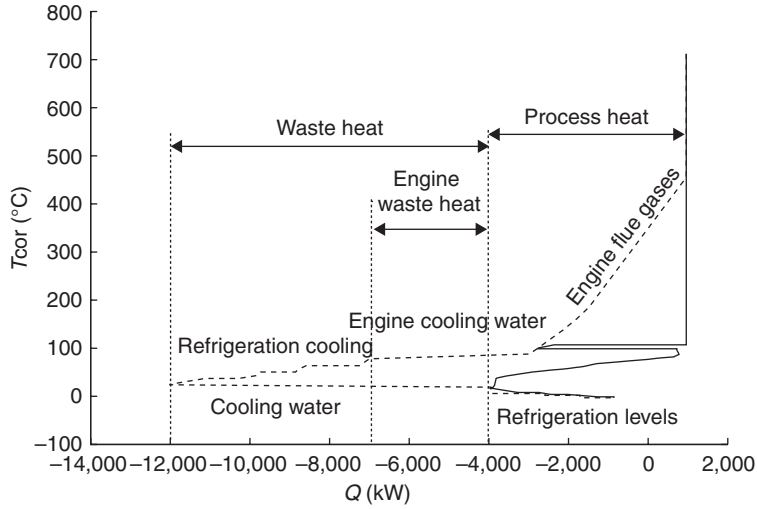
In scenarios F and G, we compare the cogeneration systems, with and without biogas production. From Figs 27.23 and 27.24, it can be seen that the system is not favourable for a cogeneration unit, mainly because the heat load of the engine cooling is only partly used to satisfy the process heat load. There is also a competition with the refrigeration cycles, which cannot be used anymore as a heat pump.

The system becomes, however, a net electricity producer. When compared with scenario A, it can calculate the marginal efficiency of the electricity production by Equation [27.13]. For scenario F, a marginal efficiency of the electricity production of 51.47% can be obtained. The efficiency can also be observed when considering the difference in the cooling water usage that increases from 3,141.6 to 8,063.3 kW (i.e. 4,921.7 kW).

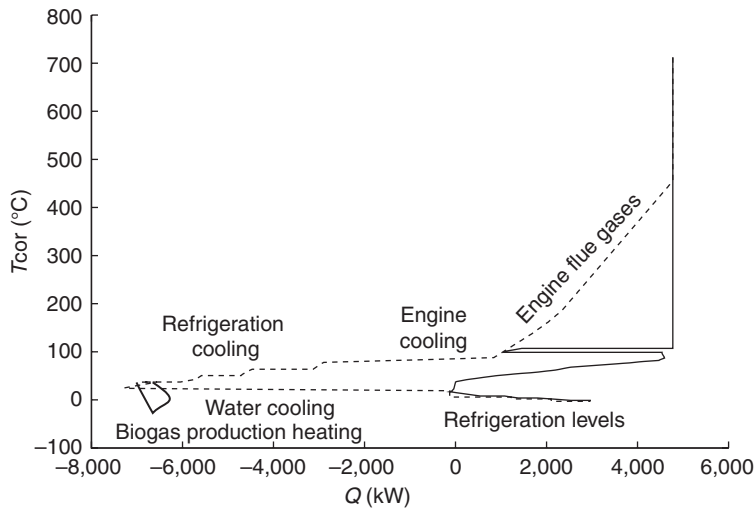
The biogas is used to reduce the amount of natural gas consumed and contributes also to the production of mechanical power.

$$\eta_{\text{marg},i} = \frac{\dot{E}_i - \dot{E}_{\text{SA}}}{\text{L}\dot{\text{H}}\text{V}_{\text{Si}} - \text{L}\dot{\text{H}}\text{V}_{\text{SA}}} \qquad [27.13]$$

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27.23 Integrated Composite Curves of energy conversion system for scenario F.



27.24 Integrated Composite Curves of energy conversion system for scenario G.

Scenarios H and I – Cogeneration, Refrigeration and Mechanical Vapour Recompression

Scenarios H and I consider, in addition, the use of mechanical vapour recompression. Changing the temperature level of the vapour recovery allows one to see the big advantage of doing the recompression. Without using the

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biogas, the solution obtained corresponds to net production of mechanical power while reducing the natural gas consumption by 28.42%. The marginal efficiency cannot be calculated anymore, except if scenario E is taken as a reference. In this case, the marginal efficiency of the electricity production is of 77.46%, much higher than could be obtained without mechanical vapour recompression.

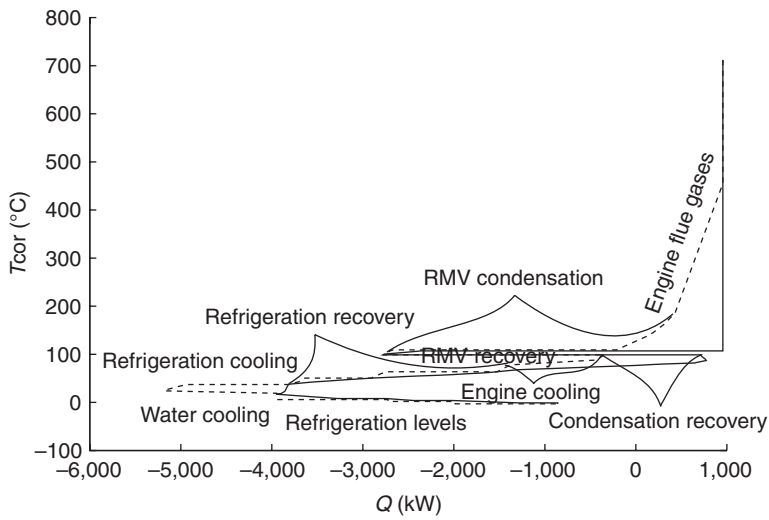
In both scenarios, mechanical vapour recompression involves 70.97% of the vapour available. The remaining vapour is directly used to heat up process streams.

From the investment point of view, the engine of scenario H is 4.6 times smaller than in scenario F without mechanical vapour recompression.

Scenarios H and I are those representing the lowest primary energy consumption. Compared with the starting point, i.e. 17,062 kW_{LHV} and 1,322 kWe corresponding to 19,422.7 kW, the primary energy saving corresponds to 84.16% of the present consumption. When considering the use of the biogas produced from the waste streams, the savings are even more attractive, with a value of 93.15%.

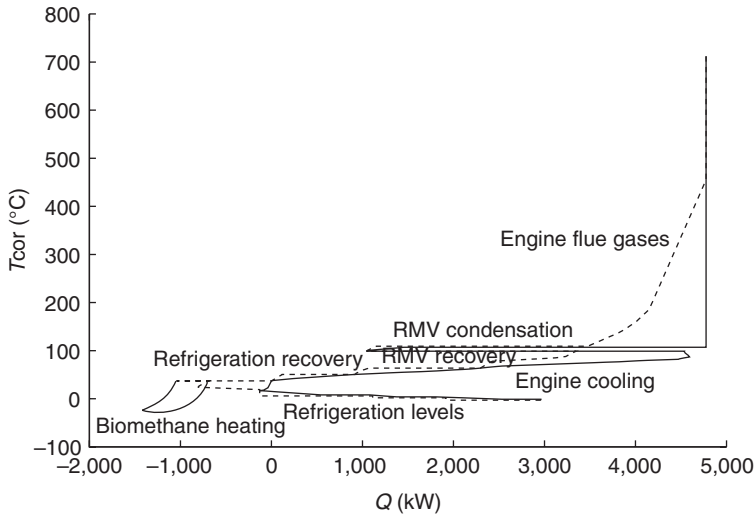
From the primary energy point of view, it should be realised that the solution with combustion and mechanical vapour recompression (scenario E) features similar results with a saving corresponding to 80.79%.

Considering the price of electricity and natural gas, it should be realised that the solution with the lowest operating cost is the one with a maximum production of electricity. The operating cost difference has, however, to compensate the investment of a bigger engine.



27.25 Integrated Composite Curves of energy conversion system for scenario H.

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27.26 Integrated Composite Curves of energy conversion system for scenario I.

The Integrated Composite Curves of the two scenarios are given on Figs 27.25 and 27.26.

27.5 Conclusion

As a result of the energy conversion system integration, the complete list of streams is defined and can be used to design the HEN design (the resulting list of hot and cold streams is given in the Annex). Before starting the HEN design, however, the list of streams has to be analysed again, and in particular the streams related to the use of hot water identified. Indeed, the needs of the units first have to be defined, in order to define the hot and cold streams requirement, once the energy target is set and the Process and Utility Pinch Points are identified; the list of streams to be considered in the HEN has to be defined by analysing the use of hot water in the system.

The first step in this analysis would be to identify the conditions of the water tanks to be considered in the system. From this first analysis the water usage network has to be designed to yield a smaller number of streams to be finally considered for the HEN design. This approach could be performed by a simple examination, or by applying mathematical programming models considering multi-period operation (Becker and Maréchal, 2012b) and combined Mass and Heat Integration (Renard *et al*., 2012).

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27.5.1 List of Streams for the HEN Design

Name	Temp: in (C)	Heat load (kW)	Temp: out (C)	DTmin/2 (K)	Unit
A	15.00	369.90	51.50	–2.00	MAK
B	48.00	188.60	65.00	–2.00	MAK
C	15.00	823.60	54.50	–2.00	MAK
D	48.00	420.10	65.00	–2.00	MAK
E	65.00	368.40	75.00	–2.00	MATMAK
F	15.00	3,450.4	80.00	–2.00	MATMAK
G	78.00	1,662.9	105.00	–2.00	MATMAK
H	105.00	3,319.8	105.00	–2.00	MATMAK
I	105.00	–17.200	100.00	10.00	RECOV
J	100.00	–3,487.9	100.00	1.00	RECOV
K	100.00	–482.80	25.00	2.00	RECOV
L	52.00	857.90	80.00	–2.00	MATMAK
M	102.50	–5,434.6	10.00	2.00	MATMAK
N	10.00	–534.80	10.00	2.00	FERM
O	10.00	–233.80	6.00	2.00	FERM
P	6.00	–425.50	6.00	2.00	FERM
Q	6.00	–714.50	1.00	2.00	FERM
R	1.00	–427.70	1.00	2.00	FERM
S	1.00	822.05	15.00	–2.00	FERM
T	15.00	4,854.0	80.00	–2.00	FERM
U	70.00	–4,858.4	5.00	2.00	FERM
V	65.00	251.90	85.00	–2.00	CIP
W	40.00	472.80	60.00	–2.00	CIP
X	15.00	1,326.9	85.00	–2.00	CIP
Y	15.00	29.300	60.00	–2.00	CIP
Z	35.00	399.34	60.00	–2.00	CIP
AA	85.00	–472.70	60.00	2.00	CIP
AB	40.00	–713.70	20.00	2.00	CIP
E1	470.00	–271.10	58.00	15.00	ENG
E1C	58.00	–89.835	40.00	15.00	ENG
E2	90.00	–280.06	80.00	2.00	ENG
BE1	87.00	–424.06	80.00	2.00	BENG
BE2	470.00	–631.71	90.00	15.00	BENG
BE3	35.00	351.22	35.00	–2.00	BENG
FRG1–	65.00	–62.913	55.00	1.85	FRG1
	65.00	–1229.2	65.00	1.16	FRG1
	190.85	–436.07	65.00	5.00	FRG1
FRG1_	1.00	1,241.6	1.00	–5.00	FRG1
FRG2–	51.67	–37.352	41.67	1.85	FRG2
	51.67	–802.01	51.67	1.16	FRG2
	166.55	–231.13	51.67	5.00	FRG2
FRG2_	–3.00	804.57	–3.00	–5.00	FRG2
FRG3–	38.33	–44.886	28.33	1.85	FRG3
	38.33	–1,042.7	38.33	1.16	FRG3
	142.18	–245.28	38.33	5.00	FRG3
FRG3_	–8.00	1,043.3	–8.00	–5.00	FRG3

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Name	Temp: in (C)	Heat load (kW)	Temp: out (C)	DTmin/2 (K)	Unit
RMV–	110.00	–4.4050	109.00	1.36	RMV
	110.00	–2,322.2	110.00	0.85	RMV
	148.56	–81.933	110.00	3.67	RMV
RMV_	98.00	2,307.2	98.00	–1.00	RMV
COOLW	17.00	674.62	22.00	–2.00	COOLW

27.6 Sources of Further Information

Useful information concerning the data collection and other application of Process Integration techniques in the food processing industry can be found in *Handbook of Water and Energy Management in Food Processing* (Klemeš *et al.*, 2008).

For further information on exergy and its application in industrial energy systems, refer to the book *Thermodynamic and Energy System Analysis* (Borel and Favrat, 2010).

In the book, *Computer Aided Process and Product* (Puigjaner and Heyen, 2006), further information can be found on computer-aided methods for process engineering, in particular concerning data reconciliation methods and the application of MILP techniques for the integration of utility systems in processes.

Most of the theoretical concepts used in this chapter are explained in detail in this handbook. However, the chapters of the UNESCO Encyclopaedia of Life Support Systems (Marechal, 2008) can also be useful. In particular, the chapter Pinch Analysis (3.19.1.7) gives a summary of the Process Integration techniques, while the chapter Improving Process Performances Using Pinch Analysis (3.19.1.8) gives a summary of how Pinch based methods can be used to improve the energy efficiency of a process and how energy conversion techniques can be integrated in an optimal way. This chapter also explains in more detail the combined use of Pinch and Exergy Analysis.

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27.8 Appendix A: Complementary Tables

Table A.1 Specific heat capacity of material streams

Material stream	Specific heat capacity – Cp (kJ/kg K)
Water	4.18
MAK	3.75
MAT	3.43
MAT + MAK	3.52
Soda (NaOH)	1.49
Sprouted barley / corn	1.8

Table A.2 Operating conditions boiler

Description	Value
LHV natural gas	47,460 (kJ/kg)
Cost natural gas	0.048 (€/kWh)
Boiler efficiency	89(%)

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Table A.3 $\Delta T_{min}/2$ for streams

Streams	$\Delta T_{min}/2$
Liquid	2
Liquid–vapour	1
Gas	4

Table A.4 Brewing operation data

Description	Value
Operation time	8,180 h/y
Daily operation time	24 h
Plant lifetime	12 y
Maintenance time	10 d/y
Failure time	5%
Interest rate	5%
Beer production (68.7 t/h)	561,000.85 t/y

Table A.5 Timeline of the batch processes in the brewery (hot part)

Stream	Process	Start time (min)	End time (min)	Total time (min)
MAK	Heating 15–65°C	0	60	60
MAT	Heating 15–65°C	0	60	60
MAT + MAK	Heating 65–75°C	60	130	70
Filtrate	Filter	130	250	120
WOK	Evaporation	250	320	70
Wort	Centrifuge	320	380	60
Wort	Cooling	380	420	40

Table A.6 Timeline of the batch processes in the brewery (cold and cleaning parts)

Stream	Process	Total time
Wort	Fermentation @ 10°C	14 d
Wort + yeast	Maturing @ 6°C	10 d
Beer chilling	Chillage @ 1°C	6–7 h
Beer	Filtration + dilution + pasteurisation	2 h
Bottle cleaning	Bottle cleaning	Continuous

27.9 Appendix B: Nomenclature

c_{el}^{+} purchase cost of electricity (MU/kJ)
c_{el}^{-} selling price of electricity (MU/kJ)
$\dot{E}_{el,process}$ electricity consumed by the process (kW)

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$\dot{E}^{+}_{\mathrm{el}}$	overall electricity purchased for the process (kW)
$\dot{E}^{-}_{\mathrm{el}}$	overall electricity exported from the process (kW)
c^{+}_{f}	fuel price (MU/kJ)
$\dot{E}^{+}_{f,u}$	nominal energy (or heating value) delivered to unit u by the fuel (e.g. natural gas) (kW)
$\dot{E}_{\mathrm{el},u}$	nominal electricity demand (+) or excess (−) of unit u (kW)
c_u	nominal operating cost per hour of unit u (excluding the fuel and electricity cost of unit u)
C_u	fixed cost related to the use of unit u
f_u	level of usage of unit u. This is a multiplication factor that is applied to all streams and flows of the unit u
$f_u^{\min}$	minimum level of usage of unit u
$f_u^{\max}$	maximum level of usage of unit u
y_u	integer variable defining the use ($y_u = 1$) or not ($y_u = 0$) of unit u
$\dot{Q}_{h/c,k,u}$	nominal heat load of hot or cold stream h/c in temperature interval k and belonging to unit u
$\dot{R}_k$	cascaded heat from the temperature interval k to the lower temperature intervals

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